THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE · MEDIUM LEVEL CHAPTER TEST SET 3

Chapter 4: **Journey Inside the Atom** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Which atom has the same number of neutrons as $^{31}_{15}\mathrm{P}$ (n = 16)? 1

(a) $^{29}_{14}\mathrm{Si}$ **(b)** $^{32}_{15}\mathrm{P}$ **(c)** $^{31}_{16}\mathrm{S}$ **(d)** $^{32}_{16}\mathrm{S}$

2. The ion Cl^- has the following number of electrons: 1

(a) 17 **(b)** 18 **(c)** 16 **(d)** 19

3. The number of shells in phosphorus (Z = 15) is: 1

(a) 4 **(b)** 5 **(c)** 2 **(d)** 3

4. The relative atomic masses of isotopes 12 u and 14 u of carbon differ by: 1

(a) 2 protons **(b)** 2 shells **(c)** 2 neutrons **(d)** 2 electrons

5. Which of these has 10 electrons? 1

(a) Cl^- **(b)** Na **(c)** K **(d)** Na^+

6. Metals generally have valence electrons: 1

(a) 1 to 3 **(b)** 0 **(c)** 8 **(d)** 5 to 7

7. An element has 2 isotopes with masses 24 and 26 in the ratio 1 : 1. The average mass is: 1

(a) 24 u **(b)** 26 u **(c)** 25 u **(d)** 50 u

8. Which two species have the same number of electrons? 1

(a) Na and Cl **(b)** Cl^- and Na^+

(c) F^- and Na^+ **(d)** Mg^{2+} and Cl^-

9. Non-metals generally have valence electrons: 1

(a) 1 to 3 **(b)** 0 **(c)** 2 only **(d)** 4 to 7

10. Isotopes of hydrogen with 1 and 2 neutrons are: 1

(a) protium and tritium **(b)** helium

(c) protium and deuterium **(d)** deuterium and tritium

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. What is meant by the term nucleon? State the number of nucleons in $^{23}_{11}\mathrm{Na}$. 2

12. Write the electronic configuration of Na^+ and Cl^-. What is the common feature? 2

13. Why is the mass of the electron neglected in calculating the mass number? 2

14. Compare the sizes of the nucleus and the atom and give the order of magnitude of each. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Write the number of protons, neutrons and electrons in Ca^{2+} formed from $^{40}_{20}\mathrm{Ca}$. 3

16. An atom X has the electronic configuration 2, 8, 7 and an atom Y has 2, 8, 2. Predict the formula of the compound and the nature of the bond. 3

17. What is the number of protons in the isobars $^{40}_{18}\mathrm{Ar}$, $^{40}_{19}\mathrm{K}$ and $^{40}_{20}\mathrm{Ca}$, and how do their neutron numbers differ? 3

OR

The nucleus of an atom has a charge of +17 and a mass of 35 u. How many protons and neutrons? What is its configuration? 3

18. Write the number of valence electrons and valency for N, F, Al, Si. 3

OR

Give two reasons why Rutherford's model had to be modified. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Explain, with two examples each, how the number of valence electrons determines the valency, the ion formed and the type of element. 5

OR

Explain the formation of the compounds MgO and NaCl in terms of electronic configurations and why MgO has a higher melting point. 5

20. Write the number of protons, neutrons and electrons, and the configuration for: $^{16}_{8}\mathrm{O}$, $^{16}_{8}\mathrm{O}^{2-}$, $^{27}_{13}\mathrm{Al}$, $^{27}_{13}\mathrm{Al}^{3+}$. 5

OR

The isotopes of an element (Z = 10) have masses 20 u, 21 u and 22 u with abundances 90 per cent, 1 per cent and 9 per cent. Find the average atomic mass and name the element. 5

END OF QUESTION PAPER — **Total: 40 Marks**

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